

Assignment: Diabetes Prediction using Neural Networks

In this assignment, you will build an **Artificial Neural Network (ANN)** using the provided **Diabetes Dataset** to predict whether a patient is likely to have diabetes based on various medical attributes.

Your goal is to implement an end-to-end **Deep Learning classification pipeline**, including data preprocessing, model building, training, evaluation, and performance analysis using TensorFlow/Keras.

This assignment will help you understand:

- End-to-end Deep Learning workflow
- Data preprocessing for neural networks
- Building Artificial Neural Networks (ANN)
- Activation functions
- Optimizers
- Model training and evaluation

Dataset Description

The dataset contains patient health-related attributes that can be used to predict the likelihood of diabetes.

Some important columns include:

Column	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Triceps skin fold thickness
Insulin	Serum insulin level
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes pedigree score
Age	Age of the patient
Outcome	Target Variable (0 = No Diabetes, 1 = Diabetes)

Assignment Tasks

You must complete the following stages.

Part 1 — Data Preparation

Prepare the dataset for training the neural network.

Tasks

1. Load the dataset using Pandas.
2. Explore the dataset:
 - a. Shape of the dataset
 - b. Column names
 - c. Data types
 - d. Missing values (if any)
3. Separate the dataset into:
 - a. Features (X)
 - b. Target Variable (y)
4. Perform Train-Test Split.
5. Apply Feature Scaling using **StandardScaler**.

Deliverables

- Dataset summary
- Feature and target separation
- Scaled training and testing datasets

Part 2 — Building the Artificial Neural Network

Create a Deep Learning model using TensorFlow/Keras.

Tasks

1. Import the required TensorFlow/Keras libraries.
2. Create a **Sequential** model.
3. Add the following layers:
 - Input Layer
 - First Hidden Layer
 - Second Hidden Layer
 - Output Layer
4. Use the following activation functions:

- ReLU for hidden layers
 - Sigmoid for the output layer
5. Compile the model using:
- Optimizer: **Adam**
 - Loss Function: **Binary Crossentropy**
 - Metric: **Accuracy**

Deliverables

- ANN architecture
- Explanation of each layer
- Activation functions used
- Optimizer and loss function selected

Part 3 — Model Training

Train the neural network using the prepared dataset.

Tasks

1. Train the model using the training dataset.
2. Select suitable values for:
 - a. Epochs

Deliverables

- Model training output
- Final training accuracy
- Number of epochs used

Part 4 — Model Evaluation

Evaluate the performance of the trained ANN.

Tasks

1. Generate predictions on the test dataset.
2. Convert prediction probabilities into binary class labels.
3. Evaluate the model using:
 - Accuracy Score
 - Confusion Matrix

Deliverables

- Model accuracy
- Confusion Matrix
- Brief interpretation of the results